

PyraSMIP User Manual

Nanopore Current Signal Acquisition Software

1. Introduction

1.1 Software Overview

PyraSMIP is a data acquisition (DAQ) and analysis software for nanopore single-molecule sensing experiments. It communicates with the PyraSMIP hardware device via a serial port, receives high-speed current signals in real time, displays live waveforms on screen, and performs baseline correction, pulse (translocation event) detection, and molecular size statistical distribution, as well as saving raw/filtered data as CSV or Parquet files.

The software includes several advanced functional modules, including waveform template matching (NCC / DTW), background noise calibration (spectral subtraction), FFT band-stop filtering, wavelet denoising, Parquet-to-CSV conversion, and automatic experiment report generation and cloud sharing based on large language models (LLM).

1.2 Key Features

- High-speed serial data acquisition (default 50 kSPS, baud rate 2,000,000)
- Real-time signal waveform display (time/current axes, with zoom and Y-axis range switching)
- Automatic baseline correction (Auto Baseline) and offset adjustment
- Pulse (molecular translocation) event detection and molecular size statistical distribution histogram
- Data recording and chunked saving (three formats: CSV / CSV ZIP / Parquet)
- Four data channel outputs: Raw, FFT (band-stop filtered), BG (background denoised), Wavelet (wavelet denoised)
- Waveform template matching detection tool (NCC / multiscale NCC / pseudo DTW / real DTW)
- FFT band-stop filter, background noise calibration, and wavelet denoising configuration windows

2. Installation & Startup

2.1 System Requirements

Item	Description
Operating System	Windows 10/11 (primary)
Hardware Interface	USB serial (device enumerated as a COM port, Windows)

3. Main Interface Overview

The main interface consists of a menu bar, a central workspace, and a bottom real-time signal area.

Area	Title	Function
Menu Bar	Menu Bar	Four menus: Export / Configuration / Tools / About
Control Panel	Control Panel	Device control, voltage control, analysis settings (scrollable on the left)
Molecular Size Distribution	DISTRIBUTION	Molecular size statistical histogram
Pulse Records	PULSE RECORDS	List of detected pulse events (Count / Type)
Real-time Signal	REAL-TIME SIGNAL	Real-time current waveform (Time vs Current)

3.1 Menu Bar

Menu	Menu Items
Export	Export Waveforms
Configuration	System Configurations
Tools	Serial Monitor, Waveform Match Tool, Background Noise Calibration, FFT Filter Configuration, Wavelet Denoising Configuration, Parquet to CSV Converter
About	Version, User ID, Help, User Manual

3.2 Control Panel

The control panel contains three groups:

- **DEVICE CONTROL:** serial port selection dropdown, plus Connect, Start, and Record buttons.
- **VOLTAGE CONTROL:** driven voltage slider (-2.0 V ~ +2.0 V), plus three filter toggles FFT Filter / Noise Filter / Wavelet.
- **ANALYSIS SETTINGS:** Pore Size, Base Line, ADC Raw, Offset slider, Auto Baseline button, Show Pulses toggle.

4. Quick Start (Typical Workflow)

The basic steps of a complete acquisition experiment are as follows:

- (1) Connect device: select the corresponding serial port in the DEVICE CONTROL dropdown and click Connect.
- (2) Set parameters: set the driven voltage in VOLTAGE CONTROL; enter the Pore Size in ANALYSIS SETTINGS.
- (3) Start sampling: click Start; the software sends a start command to the device and begins receiving data, with the live waveform scrolling in real time.
- (4) Auto baseline: after starting, Auto Baseline is automatically activated; you can also toggle it manually by clicking Auto Baseline.
- (5) Record data: click Record, set the file chunking duration in the popup dialog (default 300 seconds), then click Start Recording.
- (6) Finish & save: click Save (the original Record button), fill in experiment information and select save options and format in the popup save dialog, then click Save Data.
- (7) Stop sampling: click Stop to end the current sampling; click Disconnect to disconnect the device.

5. System Configurations

Menu Configuration → System Configurations opens the configuration window, where you can set and save/load experiment configurations.

Configuration Item	Options / Description
Solution	0.1 M KCL / 1 M KCL / 1x PBS
pH	1 ~ 12
Pore Size (nm)	Numeric input
Coating	None / 2um Au / 4um Au
PyraSMIP ID	Device number, used to determine gain (PR→1e8, UT→5e7, BG→1e8)

Click Save to save the configuration as a CSV file; Load reads an existing configuration file; after saving, the main interface pore size and gain are automatically updated.

6. Tools

6.1 Serial Monitor

Displays serial transmit/receive logs (TX/RX) in real time, including sent commands and received data byte count and rate (SPS). Used to troubleshoot device connection and communication issues. If the device has two COM ports, you can use the log to determine and switch ports.

6.2 Waveform Match Tool

Used to search a large number of CSV data files for events similar to a target template waveform, and count the number of matches.

6.2.1 Steps

- Click Load Template to select the target waveform CSV, and drag the red vertical lines in the right preview to select the template range (Dwell Start / End).
- Select the search folder (Folder); the file list refreshes automatically; you can multi-select the CSV files to search.
- Set Method (standard_ncc / multiscale_ncc / pseudo_dtw / real_dtw), Threshold (default 0.8), and sampling rate FS.
- Click Start Detection to begin detection; results appear in the Results table below (File / Match / Count / Note).

6.2.2 Matching Methods

Method	Description
standard_ncc	Standard normalized cross-correlation (single-scale template)
multiscale_ncc	Multi-scale normalized cross-correlation (template scaled 0.7~1.3x)
pseudo_dtw	Approximate matching based on sliding-window shape difference (MSE)
real_dtw	Dynamic time warping matching based on fastdtw

A threshold closer to 1.0 indicates stricter requirements on waveform shape correlation.

6.3 Background Noise Calibration

Used to capture a segment of pure background noise and build a noise spectrum for subsequent spectral subtraction denoising.

- Enter the capture duration (seconds) in Capture Duration, then click Capture Noise Profile (requires clicking Start on the main window first).
- Drag the red vertical lines in the right preview to select the noise region, then click Apply Configuration.
- The Denoising Strength (Alpha) slider (0.5 ~ 5.0, default 1.5) controls the denoising strength.

After successful application, the Noise Filter toggle on the main window is automatically enabled.

6.4 FFT Filter Configuration

Sets the lower/upper cutoff frequencies of the band-stop filter (default 45 / 55 Hz) to eliminate 50 Hz power-line interference. Click Preview Data to preview the before/after comparison, and click Apply Configuration to apply the parameters.

6.5 Wavelet Denoising Configuration

Sets the wavelet type (db4 / db8 / sym4 / sym8 / haar) and decomposition level (1 ~ 8). Click Preview Data to preview the denoising effect, and click Apply Configuration to apply. The algorithm uses wavelet soft-threshold denoising (universal threshold).

6.6 Parquet to CSV Converter

Select a Parquet file and click Convert to CSV to convert it to CSV in chunks (supports large data volumes). If the target CSV already exists, you will be asked whether to overwrite it before conversion.

7. Data File Format

Saved data files (CSV / Parquet) contain the following three columns:

Column	Unit	Description
Time (μ s)	microsecond	Sampling timestamp (relative to start time)
Current (nA)	nanoampere	Current value (converted from ADC)
Driven Voltage	V	Current driven voltage